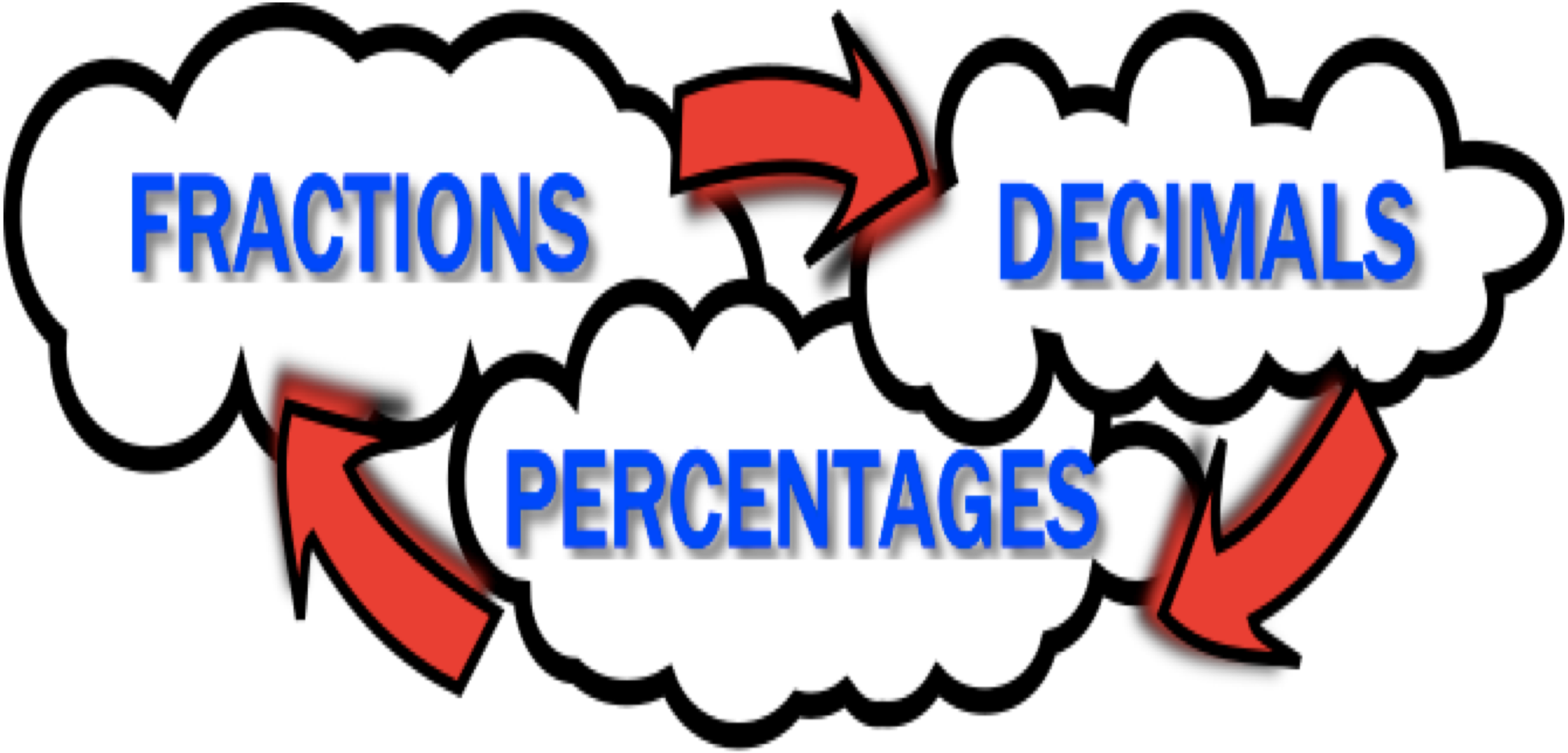
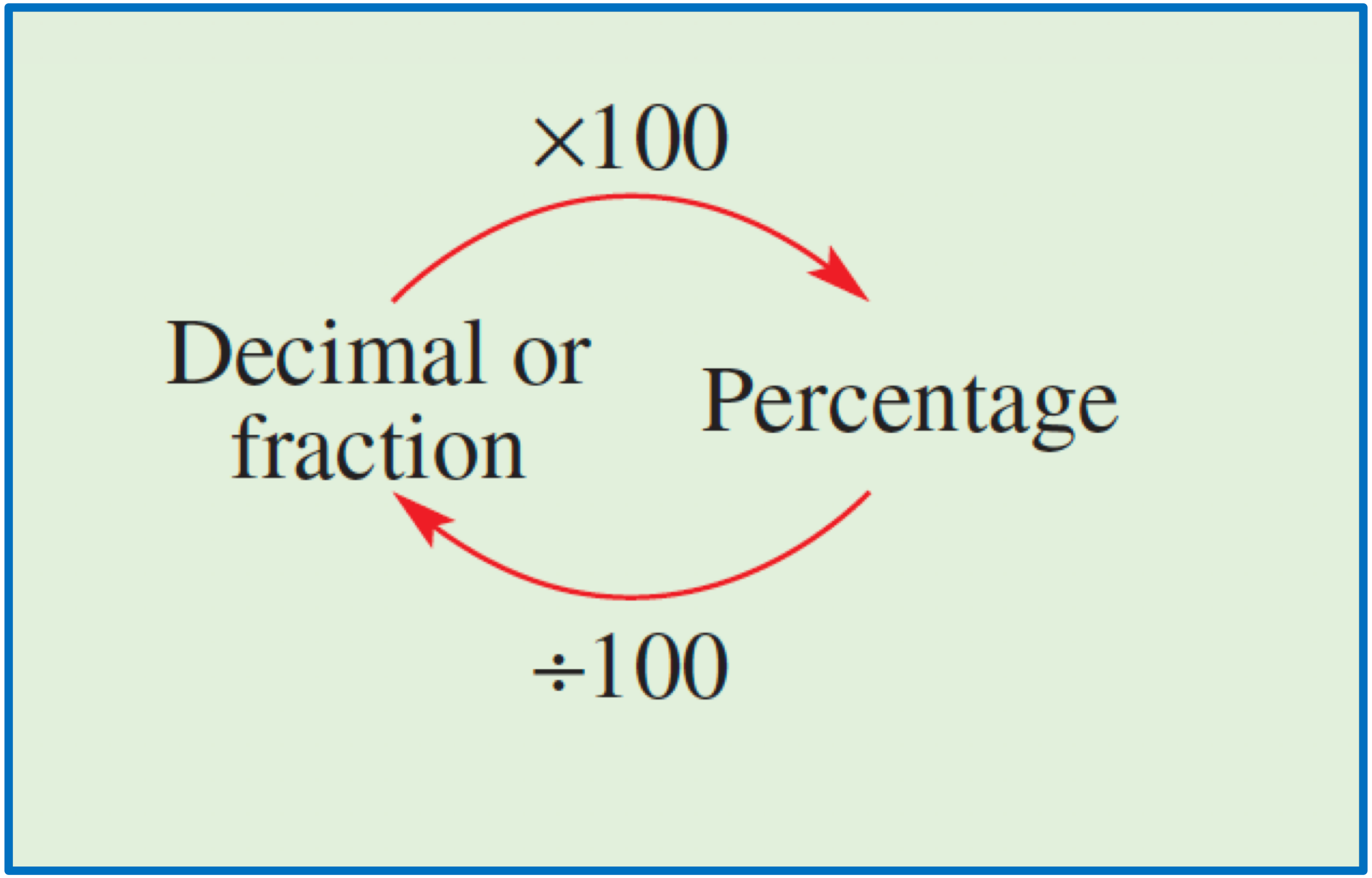
Percentages

Percentage



Converting percentages



Example

- Write the following as a percentage:
  - $\frac{12}{25} = \frac{12}{\cancel{25}} \times \cancel{100}^{4}\% = 48\%$
  - $0.375 = 0.375 \times 100\% = 37.5\%$

- Write the following as a fraction and as a decimal:
  - $52\% = \frac{\cancel{52}^{13}}{\cancel{100}_{25}} = \frac{13}{25}$ ← ALWAYS SIMPLIFY
  - $23\% = \frac{23}{100}$

Writing quantity as a percentage

- Percentage of one quantity (a) of another (b)

$$\frac{a}{b} \times 100\%$$

  - Note that a and b must be in the same unit of measurement

Example

a & b in the same unit

- Write 5g out of 50g as a percentage

a b

$\frac{\cancel{5}}{\cancel{50}_{\cancel{10}}} \times \cancel{100}^{10}\% = 10\%$

- Write 50c out of $2.50 as a percentage.

convert $2.50 to 250¢

$\frac{\cancel{50}^{1}}{\cancel{250}_{5}} \times \cancel{100}^{20}\% = 20\%$

- Write 100m out of 2km as a percentage.

2 km = 2000 m

$\frac{10\cancel{0}}{200\cancel{0}} \times 10\cancel{0}\% = 5\%$

- Find 30% of 120.

Method 1

$$120 \times 30\% = 120 \times \frac{30}{100}$$
$$= 36$$

Method 2

$$10\% \text{ of } 120 = 12$$
$$30\% \text{ of } 120 = 3 \times 12$$
$$= 36$$

Method 3

$$1\% \text{ of } 120 = 1.2$$
$$30\% \text{ of } 120 = 1.2 \times 30$$
$$= 36$$